

Interval notation



- 1
- a $(-1, 2]$ b $(-6, -3)$ c $[-3, \infty)$ d $(-3, \infty)$
e $(-\infty, 4)$
- 2
- a $(\infty, -7)$ b $(-\infty, 6]$ c $[-4, \infty)$ d $[-7, 3]$
-

Domain and range

- 3
- a
- i Domain = 8, 1, 3, 9, 2
 - ii Range = 5, 3, 4, 1, 9
 - iii Function
- c
- i Domain = 4, 8, 2, 7, 3
 - ii Range = 4, 8, 9, 1
 - iii Function
- e
- i Domain = 7, 8, 5, 3
 - ii Range = 1, 9, 3, 2, 6
 - iii Not a function
- g
- i Domain = $3 \leq x \leq 7$
 - ii Range = $5 \leq y \leq 8$
 - iii Function
- i
- i Domain = 5
 - ii Range = $2 \leq y \leq 6$
 - iii Not a function
- k
- i Domain = $-7 \leq x \leq 7$
 - ii Range = $0 \leq y \leq 7$
 - iii Function
- b
- i Domain = 2, 3, 6, 8
 - ii Range = 3, 8, 1, 7, 2
 - iii Not a function
- d
- i Domain = 1, 6, 3, 8, 2
 - ii Range = 3, 2, 7, 1
 - iii Function
- f
- i Domain = 2, 4, 6, 8, 10
 - ii Range = 2, 4, 6, 8, 9
 - iii Function
- h
- i Domain = $2 \leq x \leq 7$
 - ii Range = 5
 - iii Function
- j
- i Domain = $2 \leq x \leq 9$
 - ii Range = $3 \leq y \leq 8$
 - iii Not a function
- l
- i Domain = $-9 \leq x \leq 9$
 - ii Range = $-9 \leq y \leq 9$
 - iii Not a function

m

- i All real x
- ii $y \geq 0$
- iii Function



- i $x \geq 0$
- ii All real y
- iii Not a function

4

a

- i Domain = $(-\infty, \infty)$
- ii Range = $(-\infty, 2]$

b

- i Domain = $(-\infty, \infty)$
- ii Range = $[-2, \infty)$

c

- i Domain = $[0, 6]$
- ii Range = $[0, 7]$

5

a $(-\infty, \infty)$

b $(-\infty, \infty)$

c $[-6, 6]$

6

a False

b False

c True

d True

7

a 1

b No

c All real x

d $y > 0$

e $y = 128$

f $x = 8$

8

a 16

b $y \leq 16$

c All real x

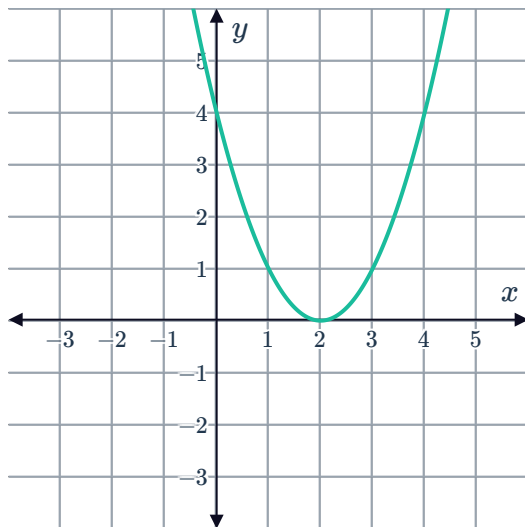
9

a $x \geq -1$

b No

10

a



b No

Continuity

11

a Yes

b

i Yes ii Yes iii No

12

a Yes

b Yes

13

a Not continuous

b Continuous

c Not continuous

d Continuous

e Continuous

f Not continuous

g Continuous

h Not continuous

14

a False

b True

c False

15

a Yes

b Yes

c Yes

16

a Yes

b Yes

17

a $(-\infty, 2) \cup (2, \infty)$

c Yes



b Yes

18

a $(0, \infty)$

b Yes

c Yes

19

a No

b $(-\infty, 2) \cup (2, \infty)$

20

a $[0, \infty)$

b $[0, \infty)$

c Yes

21

a $(-\infty, 5) \cup (5, \infty)$

b $(-\infty, 5) \cup (5, \infty)$

c Yes

22

a $(-\infty, \infty)$

b $(-\infty, \infty)$

c Yes

23

a $(-\infty, \infty)$

b $(-\infty, -5) \cup (-5, \infty)$

c No. The function is not continuous at every point in its domain.

24

a $(-\infty, \infty)$

b $(-\infty, 1) \cup (1, \infty)$

c No. The function is discontinuous at $x = 1$.



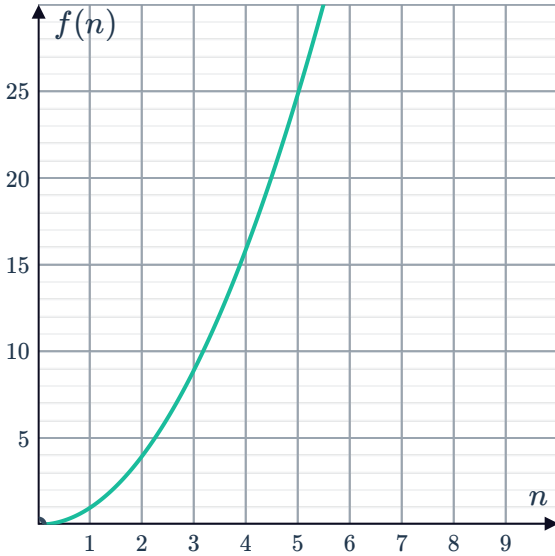
Applications

25

- a
i No ii Yes iii Yes iv Yes

b $f(n) = n^2$

c



26

- a $0 \leq x \leq 7$ b 30 c No

27

- a Yes
b The height of the water is a continuous function.
c We don't know because we don't know the location of the discontinuity.